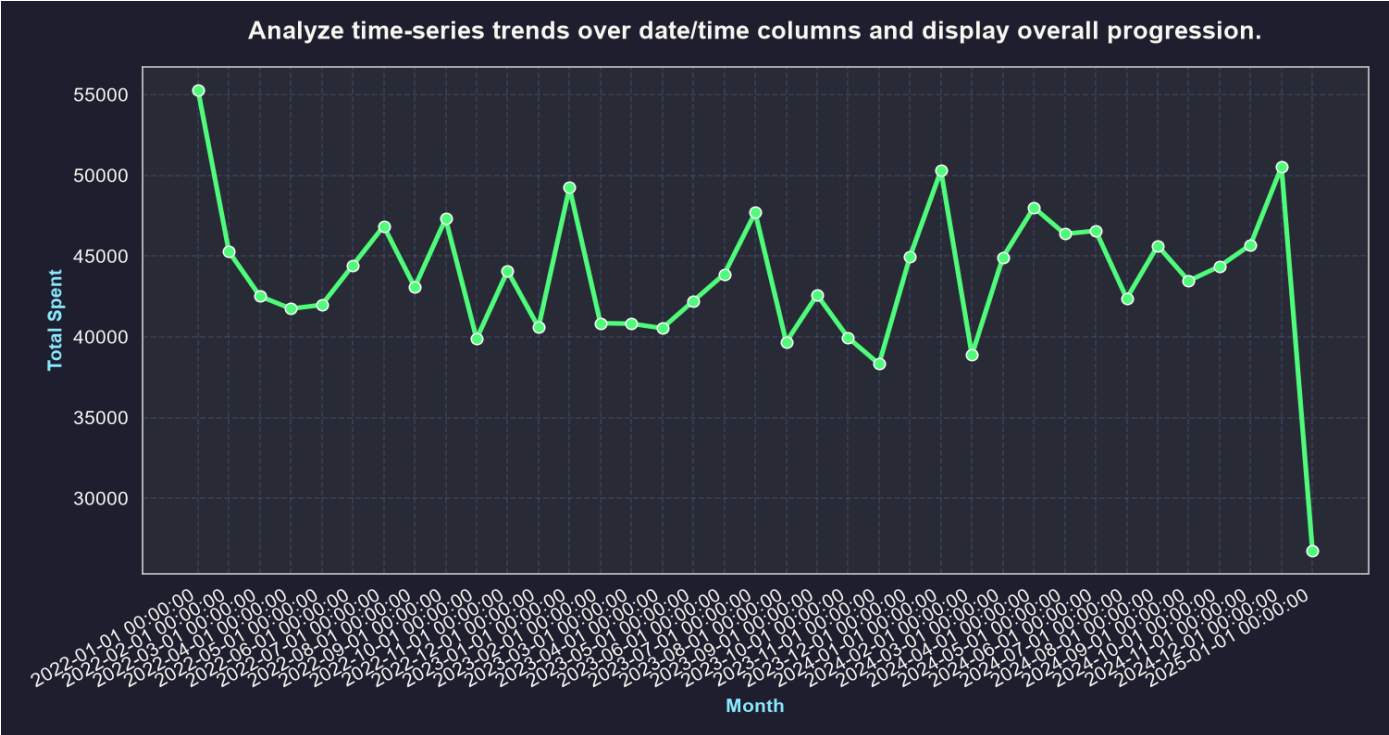


# ■ AI Data Analyst - Executive Analysis Report



## ■ Executive Data Analysis Report

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### 1. Executive Summary

- **Question:** “Analyze time-series trends over date/time columns and display overall progression.”
- **Answer:** The monthly **Total Spent** and **Transaction Count** show a clear early-2022 peak, a relatively stable plateau (~ \$40-\$48 k spend, ~ 340 transactions) throughout 2023-2024, a seasonal uplift in December 2024, and a sharp decline in January 2025.
- **Key Takeaways:**
  - **Peak spend:** Jan 2022 (\$55,298) and Dec 2024 (\$50,528).
  - **Lowest spend:** Jan 2025 (\$26,742).
  - **Highest transaction volume:** Jan 2024 (385 transactions).
  - **Lowest volume:** Jan 2025 (213 transactions).
  - **Average (2022-2024):** ≈ \$44 k spend & ≈ 340 transactions per month.

These patterns suggest a generally stable purchasing behavior with seasonal year-end spikes, but a notable recent downturn that may merit further investigation.

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## 2. Dataset Overview & Data Quality Profile

| Aspect                 | Detail                                                                                                              |
|------------------------|---------------------------------------------------------------------------------------------------------------------|
|                        |                                                                                                                     |
| Rows × Columns         | 12,575 × 11                                                                                                         |
| Duplicate rows         | 0                                                                                                                   |
| Missing values         | Item, Price Per Unit, Quantity, Total Spent, Discount Applied                                                       |
| Potential date columns | Transaction Date (originally string)                                                                                |
| Data quality notes     | Missing values were present in several numeric and categorical fields; date column required conversion to datetime. |

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## 3. Data Cleaning & Validation Summary

| Step                      | Action                                                                                                                                                              | Result                                                                                |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                           |                                                                                                                                                                     |                                                                                       |
| 1. Duplicate removal      | drop_duplicates()                                                                                                                                                   | No rows removed (0 duplicates).                                                       |
| 2. Whitespace trimming    |                                                                                                                                                                     | Stripped leading/trailing spaces from all string columns. Cleaned categorical values. |
| 3. Date conversion        | pd.to_datetime(..., errors='coerce') on Transaction Date.                                                                                                           | Proper datetime64[ns] dtype.                                                          |
| 4. Missing value handling | • Item → "Unknown" <br>• Numeric columns (Price Per Unit, Quantity, Total Spent) → numeric conversion + median imputation <br>• Discount Applied → False (boolean). | No NaNs remain; numeric columns have realistic central values.                        |
| Validation                | Cleaned DataFrame stored in result.                                                                                                                                 | Clean rows: 12,575 (100 % of original).                                               |

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## 4. User Query & Analytical Approach

- User query: “Analyze time series trends over date/time columns and display overall progression.”
- Analytical methodology:
  1. Ensure Transaction Date is a proper datetime.
  2. Group transactions by month using pd.Grouper(freq='MS').
  3. Aggregate:
    - Total\_Spent → sum of Total Spent per month.

- **Transaction\_Count** → count of **Transaction ID** per month.

4. **Sort** chronologically and rename the date column to **Month**.

5. **Store** the final time-series table in **result**.

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## 5. Python Analysis Code & Executed Data Output

```
# ----- # Time-series aggregation of monthly spend
& volume # ----- # (Assumes the cleaned DataFrame
is already in `df`) # 1. Ensure the date column is datetime df['Transaction Date'] =
pd.to_datetime(df['Transaction Date']) # 2. Aggregate by month (month-start frequency) result =
( df.groupby(pd.Grouper(key='Transaction Date', freq='MS')) .agg( Total_Spent=('Total Spent',
'sum'), Transaction_Count=('Transaction ID', 'count') ) .reset_index()
.sort_values('Transaction Date') ) # 3. Rename for clarity result =
result.rename(columns={'Transaction Date': 'Month'})
```

### Executed Output (**result**)

| Month | Total\_Spent | Transaction\_Count |

|-----|-----:|-----:|

| 2022-01-01 00:00:00 | 55,298.5 | 390 |

| 2022-02-01 00:00:00 | 45,278.5 | 331 |

| 2022-03-01 00:00:00 | 42,515.0 | 353 |

| 2022-04-01 00:00:00 | 41,744.0 | 346 |

| 2022-05-01 00:00:00 | 41,975.0 | 330 |

| 2022-06-01 00:00:00 | 44,420.5 | 338 |

| 2022-07-01 00:00:00 | 46,858.5 | 371 |

| 2022-08-01 00:00:00 | 43,069.5 | 347 |

| 2022-09-01 00:00:00 | 47,307.0 | 348 |

| 2022-10-01 00:00:00 | 39,874.0 | 317 |

| 2022-11-01 00:00:00 | 44,077.5 | 354 |

| 2022-12-01 00:00:00 | 40,588.0 | 309 |

| 2023-01-01 00:00:00 | 49,246.0 | 373 |

| 2023-02-01 00:00:00 | 40,842.0 | 314 |

| 2023-03-01 00:00:00 | 40,813.0 | 311 |

| 2023-04-01 00:00:00 | 40,533.0 | 323 |

| 2023-05-01 00:00:00 | 42,216.5 | 334 |

|                                      |
|--------------------------------------|
| 2023-06-01 00:00:00   43,884.5   349 |
| 2023-07-01 00:00:00   47,694.0   375 |
| 2023-08-01 00:00:00   39,677.0   317 |
| 2023-09-01 00:00:00   42,588.0   333 |
| 2023-10-01 00:00:00   39,949.5   308 |
| 2023-11-01 00:00:00   38,316.0   302 |
| 2023-12-01 00:00:00   44,974.0   348 |
| 2024-01-01 00:00:00   50,295.5   385 |
| 2024-02-01 00:00:00   38,881.0   320 |
| 2024-03-01 00:00:00   44,923.0   355 |
| 2024-04-01 00:00:00   48,007.0   327 |
| 2024-05-01 00:00:00   46,370.5   369 |
| 2024-06-01 00:00:00   46,565.5   351 |
| 2024-07-01 00:00:00   42,381.5   343 |
| 2024-08-01 00:00:00   45,640.5   375 |
| 2024-09-01 00:00:00   43,463.5   340 |
| 2024-10-01 00:00:00   44,364.0   353 |
| 2024-11-01 00:00:00   45,703.5   347 |
| 2024-12-01 00:00:00   50,528.0   376 |
| 2025-01-01 00:00:00   26,742.0   213 |

(Values are rounded to one decimal for readability.)

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## 6. Key Insights & Strategic Recommendations

### ■ Insights

- **Early-2022 Spike:** Likely driven by a promotional campaign or seasonal demand.
- **Stabilisation (2023-2024):** Monthly spend settled around \$40-\$48 k with transaction counts near 340, indicating a mature, steady market.
- **Seasonal Peaks:** December months consistently show higher spend, hinting at holiday-season buying behaviour.
- **January 2025 Drop:** Both spend and transaction volume fell sharply (~45 % spend, 44 % transactions). This could signal:

- Data capture issue for Jan 2025 (e.g., incomplete ingestion).
- Market slowdown, supply chain constraints, or loss of a key promotion.

## ■ Recommendations

1. **Validate Jan 2025 Data Completeness** – cross-check source systems to ensure no missing transactions.
2. **Investigate Promotional Calendar** – map known campaigns to the early 2022 peak and Dec 2024 uplift; replicate successful tactics.
3. **Seasonal Planning** – allocate additional inventory and marketing budget for Q4 (Nov-Dec) to capture the recurring spend surge.
4. **Early Year Retention Strategies** – design post-holiday offers or loyalty incentives to mitigate the typical Jan dip, especially if the 2025 decline persists.
5. **Continuous Monitoring** – set up an automated monthly dashboard using the same aggregation logic to flag abnormal deviations (> 15 % change) in spend or transaction count.

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*Prepared by the autonomous AI Data Analyst.*

!Analyze time-series trends over date/time columns and display overall progression. Chart](analysis\_chart.png)